

AE-581: Stocksolution of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ in water

Date: 2025-10-08

Tags: PNP Degassed Stocksolution
P(tBu)N(py)P(tBu) $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$
AE

Category: Two-photon

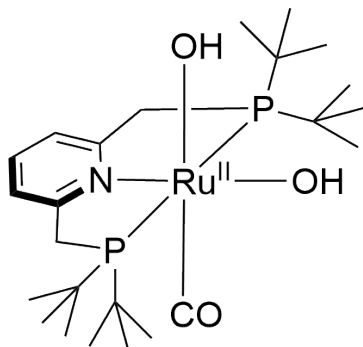
Status: Done

Created by: Alexander Eith

Objective

Making concentrated stock solution of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ in water

Reaction scheme/sample structure



Chemical Formula: $\text{C}_{24}\text{H}_{45}\text{NO}_3\text{P}_2\text{Ru}$

Molecular Weight: 558,64552

ChemDraw file: [AE-493.cdxml](#)

Literature/reference experiments

Literature	/
Reproduction	/
Related experiment	Two-photon - AE-521: Stocksolution of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ in water I

Reagents

Name	CAS/Experiment number	Amount $[\mu\text{mol}]$	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]	Density [g/mL]
$[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$	Two-photon - AE-570: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ using Ag₂O, small batch	6.3	/	/	3.52	558.65	/	/
Milli-Q water	Prep work - AE-549: Degassing of Milli-Q water	/	/	/	/	/	3.5	0.998

Procedure/observations

All steps were done according to [\[Protocol\] Schlenk Technique](#).

All steps were carried out in the dark.

Date	Time	Step	Observations	Pictures
08.10	15:20	A 5 mL Schlenk flask was set under argon.	/	/
	15:30	[Ru] was added in air	/	/
	15:45	The flask was set under argon	/	/
	15:53	To the flask water was added	yellow-orange solution	/
		The mixture was stirred at 500 rpm	no change	before filt.jpg
		A 10 mL schlenk flask was prepared	/	/
	16:07	The mixture was transferred into the new schlenk flask and thereby filtered thorough a PA syringe filter (pore diamter = 0.2 µm)	significatnly lighter in tone, hue did not change	after filt.jpg
	16:10	The flask was stored at -20 °C in the dark under argon in Equipment - Freezer Lab E004 CEEC II	AE-581	/

Product characterization

Sample	Volume [mL]	Purity	Conecntration [mM]	Description	Photo	Storage location
AE-581	3.5	98 %	concentrated solution (aprox. 1.44)	light yellow solution		Equipment - Freezer Lab E004 CEEC II

Result

Worked as expected

Linked experiments

Prep work - [AE-549: Degassing of Milli-Q water](#)

Two-photon - [AE-499: Stocksolution of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] in water](#)

Two-photon - [AE-521: Stocksolution of \[Ru\(CO\)\(OH\)2\(PNP{tBu}\)\] in water I](#)

Two-photon - [AE-570: Synthesis of \[Ru\(CO\)\(OH\)2\(PNPtBu\)\] using Ag2O, small batch](#)

Linked resources

Equipment - [Freezer Lab E004 CEEC II](#)

Protocol - [Schlenk Technique](#)

Attached files

after filt.jpg

sha256: e18755d44ca32b0147654a4c2e533f1dcc53ccaa528bdfd67dd705c54c119b9f



before filt.jpg

sha256: 912b93a5af1aca32f21f469705f9c05046492688017fbbfb8bbc792ddfff267d



AE-493.cdxml

sha256: df124f38dd8ca112a9ed9e47dc25758df4cf442a4bb657f1821b70595f507791



Unique eLabID: 20251008-752b93d19f83ae2ac6b67368179ecedabfc62162
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=3134>